

RESULTS DOSSIER

Isotope results and calculation certificate

Report number
RES-20261005-63C68087

Issue date
5 October 2026

Natan Pereira BTS coral series

Client: Natan Pereira
Project: BTS coral
Method: DEMO carbonate method / v4
Qualification: 0c8457d7fe6b47939ec1a973d4a3a0dd

Result status: blocked

This calculation dossier is a release certificate only for rows explicitly marked released. Blocked and review results remain provisional. Missing uncertainties are not zero.
Simulation assumption: externally referenced isotope means are retained, with no second normalization. The displayed current mock qualification is a comparison model, not evidence of the original historical Qtegra/ISODAT calibration. Qtegra drift correction is disabled.

SIMULATED QUALIFICATION. See the observed/synthetic origin and calibration verification of each result.

Results and expanded uncertainty / per mille VPDB

G1: Natan Pereira BTS coral series 1

Sample / analysis	Group	Carbon-13	U (C)	Oxygen-18	U (O)	Decision
BTS / 2 1003*	G1	-2.1548	0.0745	-3.5874	0.1157	blocked
BTS / 3 642*	G1	-1.7885	0.0730	-3.5065	0.1158	blocked
BTS / 4 384*	G1	—	—	—	—	blocked
BTS / 5 634	G1	—	—	—	—	blocked
BTS / 7 632	G1	-1.5215	0.0721	-3.5006	0.1159	blocked
BTS / 8 951	G1	—	—	—	—	blocked
BTS / 9 405	G1	-1.8275	0.0732	-3.5983	0.1157	blocked
BTS / 10 872	G1	-1.4272	0.0718	-3.6318	0.1156	blocked
BTS / 12 952	G1	—	—	—	—	blocked

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Sample / analysis	Group	Carbon-13	U (C)	Oxygen-18	U (O)	Decision
BTS / 13 989	G1	—	—	—	—	blocked
BTS / 14 873	G1	-1.7011	0.0727	-3.3842	0.1161	blocked
BTS / 15 893	G1	—	—	—	—	blocked
BTS / 17 953	G1	—	—	—	—	blocked
BTS / 18 371	G1	—	—	—	—	blocked
BTS / 19 992	G1	—	—	—	—	blocked
BTS / 20 1001	G1	—	—	—	—	blocked
BTS / 22 641	G1	-1.6334	0.0725	-3.5558	0.1157	blocked
BTS / 23 1002	G1	-2.4424	0.0758	-3.0229	0.1169	blocked
BTS / 24 401	G1	-2.1476	0.0745	-3.9366	0.1150	blocked
BTS / 25 879	G1	-1.2003	0.0711	-3.6754	0.1155	blocked
BTS / 27 393	G1	-2.0035	0.0739	-3.9366	0.1150	blocked
BTS / 28 389	G1	-3.0565	0.0790	-4.0142	0.1149	blocked
BTS / 29 883	G1	—	—	—	—	blocked
BTS / 30 919	G1	—	—	—	—	blocked
BTS / 32 875	G1	—	—	—	—	blocked
BTS / 33 861	G1	-1.7300	0.0728	-3.5214	0.1158	blocked
BTS / 34 942	G1	—	—	—	—	blocked

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Sample / analysis	Group	Carbon-13	U (C)	Oxygen-18	U (O)	Decision
BTS / 35 976	G1	—	—	—	—	blocked
BTS / 37 399	G1	-3.0085	0.0787	-3.4822	0.1159	blocked
BTS / 38 971	G1	—	—	—	—	blocked
BTS / 39 644	G1	-1.9688	0.0737	-2.4878	0.1181	blocked
BTS / 40 965	G1	—	—	—	—	blocked
BTS / 42 969	G1	—	—	—	—	blocked
BTS / 43 390	G1	—	—	—	—	blocked
BTS / 44 637	G1	-2.5433	0.0763	-3.8030	0.1153	blocked
BTS / 45 870	G1	—	—	—	—	blocked

Calculation model and uncertainty

$z = x - c(p - p_0)$; $y = A1 + (z - M1)(A2 - A1)/(M2 - M1)$. Already normalized Qtegra values retain the documented normalization. Only remaining validated corrections are applied.

$u_{\text{norm}}^2 = J^T \Sigma_{\text{anchors}} J$. $u_c^2 = u_{\text{prec}}^2 + u_{\text{norm}}^2 + u_{\text{corr}}^2 + \sum(u_j^2)$, for independent components; $U = k \times u_c$. u_{prec} is an individual QC standard deviation, not the standard error of its mean. Correction sensitivity includes its shared contribution to sample and anchors.

Evaluation 035af3e5975e

Carbon-13: slope 1; intercept -0.12. Assigned anchors: [-5.014, 1.95]. Measured means: [-4.893999301869521, 2.069999886722692].

Anchor covariance	A1	A2	M1	M2
A1	0.001225	0	0	0
A2	0	0.0004	0	0
M1	0	0	4.2124e-06	0
M2	0	0	0	4.2128e-06

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Residual correction: Residual intensity effect. Predictor: i44_v; slope c=0.008777, u(c)=0.00017862; reference p0=6; validated domain 3.55 to 8.45.

Carbon-13 / sample	u_prec	u_norm	u_corr	u_c	k	U
BTS / 2 1003*	0.0299	0.0223	0.0002	0.0373	2.0000	0.0745
BTS / 3 642*	0.0299	0.0210	0.0000	0.0365	2.0000	0.0730
BTS / 4 384*	—	—	—	—	—	—
BTS / 5 634	—	—	—	—	—	—
BTS / 7 632	0.0299	0.0202	0.0001	0.0361	2.0000	0.0721
BTS / 8 951	—	—	—	—	—	—
BTS / 9 405	0.0299	0.0211	0.0003	0.0366	2.0000	0.0732
BTS / 10 872	0.0299	0.0199	0.0001	0.0359	2.0000	0.0718
BTS / 12 952	—	—	—	—	—	—
BTS / 13 989	—	—	—	—	—	—
BTS / 14 873	0.0299	0.0207	0.0002	0.0364	2.0000	0.0727
BTS / 15 893	—	—	—	—	—	—
BTS / 17 953	—	—	—	—	—	—
BTS / 18 371	—	—	—	—	—	—
BTS / 19 992	—	—	—	—	—	—
BTS / 20 1001	—	—	—	—	—	—
BTS / 22 641	0.0299	0.0205	0.0000	0.0362	2.0000	0.0725

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Carbon-13 / sample	u_prec	u_norm	u_corr	u_c	k	U
BTS / 23 1002	0.0299	0.0233	0.0004	0.0379	2.0000	0.0758
BTS / 24 401	0.0299	0.0222	0.0001	0.0372	2.0000	0.0745
BTS / 25 879	0.0299	0.0193	0.0003	0.0356	2.0000	0.0711
BTS / 27 393	0.0299	0.0217	0.0001	0.0369	2.0000	0.0739
BTS / 28 389	0.0299	0.0258	0.0001	0.0395	2.0000	0.0790
BTS / 29 883	—	—	—	—	—	—
BTS / 30 919	—	—	—	—	—	—
BTS / 32 875	—	—	—	—	—	—
BTS / 33 861	0.0299	0.0208	0.0001	0.0364	2.0000	0.0728
BTS / 34 942	—	—	—	—	—	—
BTS / 35 976	—	—	—	—	—	—
BTS / 37 399	0.0299	0.0256	0.0001	0.0394	2.0000	0.0787
BTS / 38 971	—	—	—	—	—	—
BTS / 39 644	0.0299	0.0216	0.0000	0.0369	2.0000	0.0737
BTS / 40 965	—	—	—	—	—	—
BTS / 42 969	—	—	—	—	—	—
BTS / 43 390	—	—	—	—	—	—
BTS / 44 637	0.0299	0.0237	0.0003	0.0381	2.0000	0.0763

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Carbon-13 / sample	u_prec	u_norm	u_corr	u_c	k	U
BTS / 45 870	—	—	—	—	—	—

precision: Reviewed period DEMO controlled period 3 (2026-08-03); SD of individual results, not standard error. Failed runs excluded with their evidence retained.

normalization: Propagated from both assigned values and measured anchor means

secondary_correction: Separate synthetic training and validation experiments, different materials and independent random draws; neither shares carousel anchor or routine QC observations.

Oxygen-18: slope 1; intercept 0.15. Assigned anchors: [-23.01, -2.2]. Measured means: [-23.159999945696633, -2.35000000881116].

Anchor covariance	A1	A2	M1	M2
A1	0.0049	0	0	0
A2	0	0.0016	0	0
M1	0	0	8.8304e-06	0
M2	0	0	0	8.8305e-06

Residual correction: Residual intensity effect. Predictor: i44_v; slope c=0.014983, u(c)=0.00039616; reference p0=6; validated domain 3.55 to 8.45.

Oxygen-18 / sample	u_prec	u_norm	u_corr	u_c	k	U
BTS / 2 1003*	0.0438	0.0377	0.0004	0.0578	2.0000	0.1157
BTS / 3 642*	0.0438	0.0378	0.0000	0.0579	2.0000	0.1158
BTS / 4 384*	—	—	—	—	—	—
BTS / 5 634	—	—	—	—	—	—
BTS / 7 632	0.0438	0.0379	0.0003	0.0579	2.0000	0.1159
BTS / 8 951	—	—	—	—	—	—
BTS / 9 405	0.0438	0.0377	0.0006	0.0578	2.0000	0.1157
BTS / 10 872	0.0438	0.0377	0.0001	0.0578	2.0000	0.1156
BTS / 12 952	—	—	—	—	—	—

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Oxygen-18 / sample	u_prec	u_norm	u_corr	u_c	k	U
BTS / 13 989	—	—	—	—	—	—
BTS / 14 873	0.0438	0.0380	0.0005	0.0580	2.0000	0.1161
BTS / 15 893	—	—	—	—	—	—
BTS / 17 953	—	—	—	—	—	—
BTS / 18 371	—	—	—	—	—	—
BTS / 19 992	—	—	—	—	—	—
BTS / 20 1001	—	—	—	—	—	—
BTS / 22 641	0.0438	0.0378	0.0000	0.0579	2.0000	0.1157
BTS / 23 1002	0.0438	0.0386	0.0008	0.0584	2.0000	0.1169
BTS / 24 401	0.0438	0.0372	0.0003	0.0575	2.0000	0.1150
BTS / 25 879	0.0438	0.0376	0.0007	0.0578	2.0000	0.1155
BTS / 27 393	0.0438	0.0372	0.0003	0.0575	2.0000	0.1150
BTS / 28 389	0.0438	0.0371	0.0003	0.0574	2.0000	0.1149
BTS / 29 883	—	—	—	—	—	—
BTS / 30 919	—	—	—	—	—	—
BTS / 32 875	—	—	—	—	—	—
BTS / 33 861	0.0438	0.0378	0.0002	0.0579	2.0000	0.1158
BTS / 34 942	—	—	—	—	—	—

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Oxygen-18 / sample	u_prec	u_norm	u_corr	u_c	k	U
BTS / 35 976	—	—	—	—	—	—
BTS / 37 399	0.0438	0.0379	0.0001	0.0579	2.0000	0.1159
BTS / 38 971	—	—	—	—	—	—
BTS / 39 644	0.0438	0.0396	0.0000	0.0591	2.0000	0.1181
BTS / 40 965	—	—	—	—	—	—
BTS / 42 969	—	—	—	—	—	—
BTS / 43 390	—	—	—	—	—	—
BTS / 44 637	0.0438	0.0374	0.0006	0.0576	2.0000	0.1153
BTS / 45 870	—	—	—	—	—	—

precision: Reviewed period DEMO controlled period 3 (2026-08-03); SD of individual results, not standard error. Failed runs excluded with their evidence retained.

normalization: Propagated from both assigned values and measured anchor means

secondary_correction: Separate synthetic training and validation experiments, different materials and independent random draws; neither shares carousel anchor or routine QC observations.

QC correction validation

Run / isotope	Paired QC	SD before	SD after	SD reduction	Status
4c0283d537ed / d13c	7	0.1181	0.1192	-1.0%	not_improved
4c0283d537ed / d18o	7	0.2788	0.2768	0.7%	review_required

Traceability and review

Source / evaluation	Original SHA-256	Origin / calibration transfer
R1 / 4c0283d537ed46c2a2a0bf83e8a4170f Evaluation: 035af3e5975e4b52bd758a1da54bab 5c	cf84c933e494ca4a89622328b74302cbc4317f42 266330bdaef4ef5370369dd5	observed / simulation_assumption

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Review code	Issue requiring review
E1	mass_ug missing or outside validated range
E2	pressure_mismatch_v missing or outside validated range
E3	Historical calibration transfer is unverified. Simulated qualification supports workflow review only.
E4	d13c independent QC did not pass precision, bias and individual-analysis criteria
E5	d18o independent QC did not pass precision, bias and individual-analysis criteria
E6	Historical QC control signals require investigation and a reviewed instrument state
E7	Acquisition predates method approval/activation
E8	retrospective data remain review-only
E9	No valid periodic qualification covered the acquisition time
E10	Otegra pressure adjustment failed
E11	i44_v missing or outside validated range
E12	d13c internal SD missing or at/above limit
E13	d13c correction predictor missing or outside validated domain
E14	d18o internal SD missing or at/above limit
E15	d18o correction predictor missing or outside validated domain
E16	d13c: Correction predictor is missing or outside its validated domain
E17	d18o: Correction predictor is missing or outside its validated domain

Sample / analysis	Source	Decision	Review codes / accepted exceptions
BTS / 2 1003*	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 3 642*	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 4 384*	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 5 634	R1	blocked	E1, E11, E2, E13, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 7 632	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 8 951	R1	blocked	E1, E11, E2, E12, E13, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9

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Sample / analysis	Source	Decision	Review codes / accepted exceptions
BTS / 9 405	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 10 872	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 12 952	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 13 989	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 14 873	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 15 893	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 17 953	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 18 371	R1	blocked	E1, E11, E2, E12, E13, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 19 992	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 20 1001	R1	blocked	E1, E11, E2, E12, E13, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 22 641	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 23 1002	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 24 401	R1	blocked	E1, E3, E4, E5, E6, E7, E8, E9
BTS / 25 879	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 27 393	R1	blocked	E1, E3, E4, E5, E6, E7, E8, E9
BTS / 28 389	R1	blocked	E1, E12, E3, E4, E5, E6, E7, E8, E9
BTS / 29 883	R1	blocked	E1, E11, E2, E13, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 30 919	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9

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BTS / 32 875	R1	blocked	E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 33 861	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 34 942	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 35 976	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 37 399	R1	blocked	E1, E3, E4, E5, E6, E7, E8, E9
BTS / 38 971	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 39 644	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 40 965	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 42 969	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 43 390	R1	blocked	E1, E11, E2, E13, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9
BTS / 44 637	R1	blocked	E1, E2, E3, E4, E5, E6, E7, E8, E9
BTS / 45 870	R1	blocked	E10, E1, E11, E2, E12, E13, E14, E15, E16, E17, E3, E4, E5, E6, E7, E8, E9

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